THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING: LEVEL I EXAMINATION

DEPARTMENT OF CIVIL ENGINEERING

CIVL1105: Environmental Engineering

Dec 13, 2023 TIME: 09:30 AM – 12:30 PM

Answer ALL questions. Section A carries 30 marks, and Section B carries 70 marks.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Section A. (Ten One-choice questions with 3 marks each, no deduction of marks for each wrong answer.)

You may need to use Appendix A at the end of the exam paper for your calculations.

1. The average concentration of sulfur dioxide (SO_2) in the air is measured as 11.5 ppb in the US in 2022. Under the same conditions, what is the above SO_2 concentration expressed in $\mu g/m^3$? Given that the gas has a temperature of 25 °C and a pressure of 1 atm.
   (a) 29.7; (b) 30.1; (c) 32.4; or (d) 32.8

2. If 2100 kJ of energy is just enough to melt 3.0 kg of a substance, what is the substance's latent heat of fusion?
   (a) 6300 kJ·kg; (b) 6300 kJ/kg; (c) 700 kJ·kg; or (d) 700 kJ/kg

3. The standard enthalpies of the formation of $CH_4(g)$, $CO_2(g)$, and $H_2O(l)$ are −74.9, −393.5, and −241.8 kJ/mol, respectively. What is the amount of heat evolved by burning 1 m^3 of methane under conditions of 25 °C and 1 atm? Given that the standard enthalpy of $O_2(g)$ is zero.
   (a) 802 kJ; (b) 3110 kJ; (c) 32743 kJ; or (d) 35813 kJ

4. Which of the following statements regarding water distribution is **NOT** true?
   (a) Groundwater, glaciers, and ice caps belong to the freshwater, which occupies around 2.5% of the earth's total water.
   (b) Surface water is any body of water above ground.
   (c) The ocean, which occupies around 96.5% of the earth's total water, is also considered as surface water.
   (d) The surface fresh water only occupies 1.3% of the earth's water.

5. Which of the following is **NOT** true for the description of eutrophication?
   (a) It is mainly caused by inorganic plant nutrients such as N and P.
   (b) The rule of thumb on limiting nutrients, when N/P > 10, it should be the P limiting.
   (c) A eutrophication lake, it is high in nutrients, high turbid; but low DO and low diversity.
   (d) Eutrophication occurs by humans.

6. Which of the following statements regarding disinfection is **NOT** true?
   (a) The objective of disinfection is to kill pathogens in water.
   (b) The free chlorine, (including Cl_2, NaOCl, and $Ca(OCl)_2$) is one of the most common disinfectants.
   (c) Disinfection at slightly acidic pH, the disinfection power of HOCl is much higher than that of OCl^-.
   (d) The disinfection byproduct is one of the disadvantages of water treatment.

7. Which of the following membrane technologies is **NOT** suitable for metal ion separation?
   (a) Ion exchange membranes
   (b) Nanofiltration membranes.
   (c) Microfiltration membranes.
   (d) Reverse osmosis membranes.

8. There is a small city of 500,000 that sends 0.6 m^3 of sewage /person/d to the wastewater treatment plant. A circular primary clarifier would be designed to have an average detention time of 0.96 hours and an average overflow rate of 100 m/d. What would be the dimensions (the clarifier surface area, tank diameter, and tank depth) of this clarifier?
(a) 3000 m^2, 62 m, 4.0 m.
(b) 3000 m^2, 31 m, 96 m.
(c) 375 m^2, 22 m, 3.3 m.
(d) 375 m^2, 22 m, 96 m.

9. The stratospheric ozone layer is vital for absorbing harmful ultraviolet radiation from the sun. Certain compounds have the potential to degrade this layer, leading to its depletion. Among the compounds listed below, which one has the minimal impact on the depletion of ozone in the stratosphere?
(a) $CFCl_3$; (b) CHF_2Cl; (c) CH_2FCF_3; or (d) CF_2ClBr

10. Greenhouse gases play a pivotal role in regulating the Earth's temperature by trapping heat in the atmosphere. Among the compounds presented below, which one is **NOT** typically recognized as contributing to this greenhouse effect?
(a) N_2; (b) CO_2; (c) CCl_4; or (d) H_2O

Section B. (Four questions with unequal marks)

You may need to use Appendix A at the end of the exam paper for your calculations.

11. (a) Suppose the total world energy consumption of fossil fuels, equal to 9.2×10^{16} kJ/yr, were to be obtained entirely by combustion of petroleum with the approximate chemical formula C_2H_3.

    i. Please find out the balanced chemical reaction of C_2H_3;

    (4 marks)

    ii. Given that the combustion of petroleum releases about 45×10^3 kJ/kg. Please estimate the emissions of CO_2 per year in the unit of kg.

    (8 marks)

    (b) Water is frequently disinfected with chlorine gas, forming hypochlorous acid (HOCl), which partially ionizes to hypochlorite and hydrogen ions as follows:

    $$HOCl = H^+ + OCl^-$$ with equilibrium constant $K_{a1} = 2.9\times10^{-8}$

    The amount of $[HOCl]$, which is the desired disinfectant, depends on the pH. Please find the fraction that is hypochlorous acid—that is, $[HOCl] / \{[HOCl] + [OCl^-]\}$—as a function of pH. What would the hypochlorous fraction (in percentage) be for pH = 5, 7, and 9?

    (8 marks)

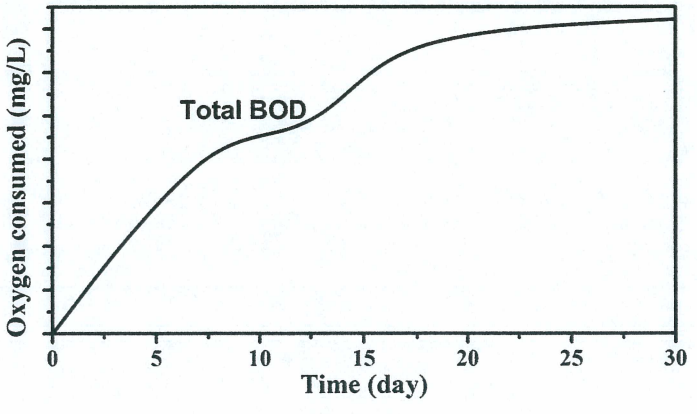
12. (a) The following picture illustrates the carbonaceous and nitrogenous biochemical oxygen demand. Please complete the following picture:

    i. Please draw out the CBOD, NBOD, and the remaining OD curves in the following picture;

    ii. Please point out the BOD_5 in this picture;

    iii. Please explain CBOD, NBOD, total BOD, and the ultimate carbonaceous oxygen demand L_0.

    (8 marks)



(b) The dilution factor for an unseeded mixture of wastewater and pure water is 0.08. The DO of the mixture was initially 10.0 mg/L, and 5 days later, it dropped 8.0 mg/L. The reaction rate constant k has been found to be 0.22 day^{-1}. What would be the ultimate CBOD?

(8 marks)

13. (a) Compared to the classic methods (including sedimentation, coagulation and flocculation, filtration disinfection, and water softening), please provide your ideas (advantages and disadvantages) of membrane technology in water treatment. (**no more than 500 words**).

(8 marks)

(b) An inland city is close to Salt Lake, and its groundwater source with a hardness of 300 mg/L as Ca^{2+}. If people want to use the groundwater for drinking purposes, please design an appropriate treatment process, and explain your design purposes.

(8 marks)

(c) A 600 m^3 volume rectangular shows a well-mixed aeration lagoon. It receives 12.5 m^3/h of wastewater with a BOD_5 of 336 mg/L for treatment. The biodegradation rate constants for the wastewater are: endogenous decay constant $k_d = 0.08\ d^{-1}$, maximum specific microbial growth rate $\mu_m = 1.10\ d^{-1}$, and the half velocity constant $K_s = 76.0$ mg BOD_5/L. What is the efficiency of the lagoon?

(8 marks)

14. Motor vehicles, such as cars, trucks, buses, and motorcycles, emit various pollutants that can negatively impact the environment and human health.
(a) SO_2 and NO_2 are commonly emitted from vehicles. Please RESPECTIVELY explain the environmental and health impacts of these pollutants.

(4 marks)

(b) Vehicle emissions include particulate matter (PM), which comprises tiny particles and liquid droplets in the air. Explain why $PM_{2.5}$ is considered more hazardous than PM_{10} in terms of their potential health risks.

(2 marks)

(c) Suggest at least two strategies or methods to mitigate the release of pollutants from vehicles.

(4 marks)

Appendix A. Information for Reference

I. Atomic weights

H 1; C 12; N 14; O 16; F 19; Na 23; Mg 24.3; Al 27; P 31; S 32; Cl 35.5; K39; Ca 40; Cr 52; Mn 55; Fe 55.8; Ni 58.7; Cu 63.5; Zn 65.4; I 127.

II. Properties of water

Density = 1000 kg/m^3
Viscosity (15 °C) = 1.1404 mPa·s
Specific heat = 4.18 kJ/(kg·°C)
Heat of vaporization (15 °C) = 2465 kJ/kg
Heat of fusion = 333 kJ/kg

III. Other constants

Ideal gas law constant = 0.082056 L·atm/(K·mol)
Partial pressure of CO_2 in atmosphere: $1\times10^{-3.5}$ atm
Henry's Law constant for CO_2: 0.03162 mol/(L·atm)
Acidity constants for H_2CO_3: $pK_1 = 6.3$; $pK_2 = 10.3$
Solubility of $CaCO_3$(s): $pK_{so} = 8.3$
Henry's Law constant for O_2 (25 °C) = 0.001263 mol/(L·atm)
Dissociation constant of water (25 °C) $K_w = 1\times10^{-14}$
Equivalent Weight (EW) for Ca^{2+} = 20 mg/meq
EW for Mg^{2+} = 12.2 mg/meq
EW for $CaCO_3$ = 50 mg/meq

- END OF PAPER -